Waveforms and Intro to Spectra

Intro to Audio Tech 1
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Monday, September 8

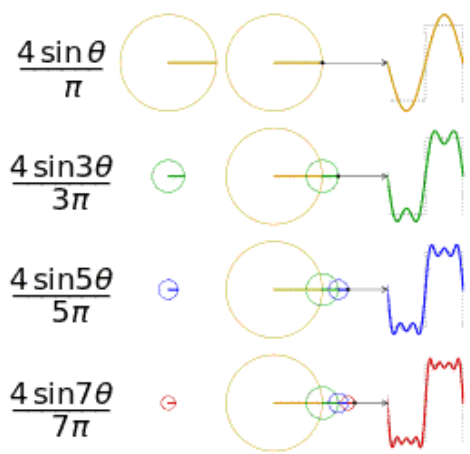
Georgia Tech

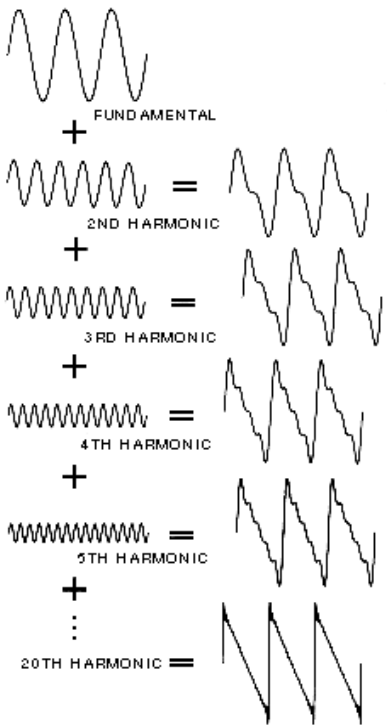
Lesson Outline

- Additive Synthesis
- Classical Waveforms
- Up Next…

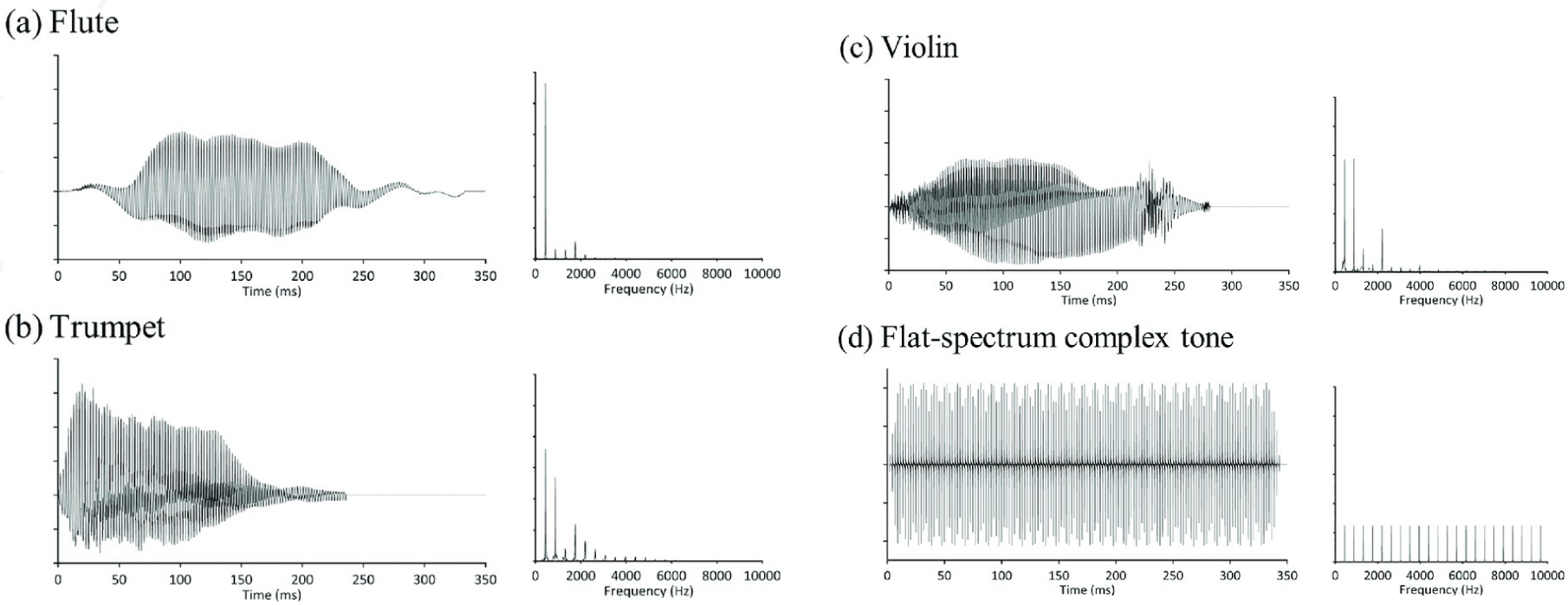
Remember: Sinusoids and Complex Tones

- Sine/cosine waves have a single frequency
- In the real world, almost all sounds are made up of multiple frequencies
- We can decompose a periodic signal as a sum of sine waves!
- Sinusoids are the building blocks of sound





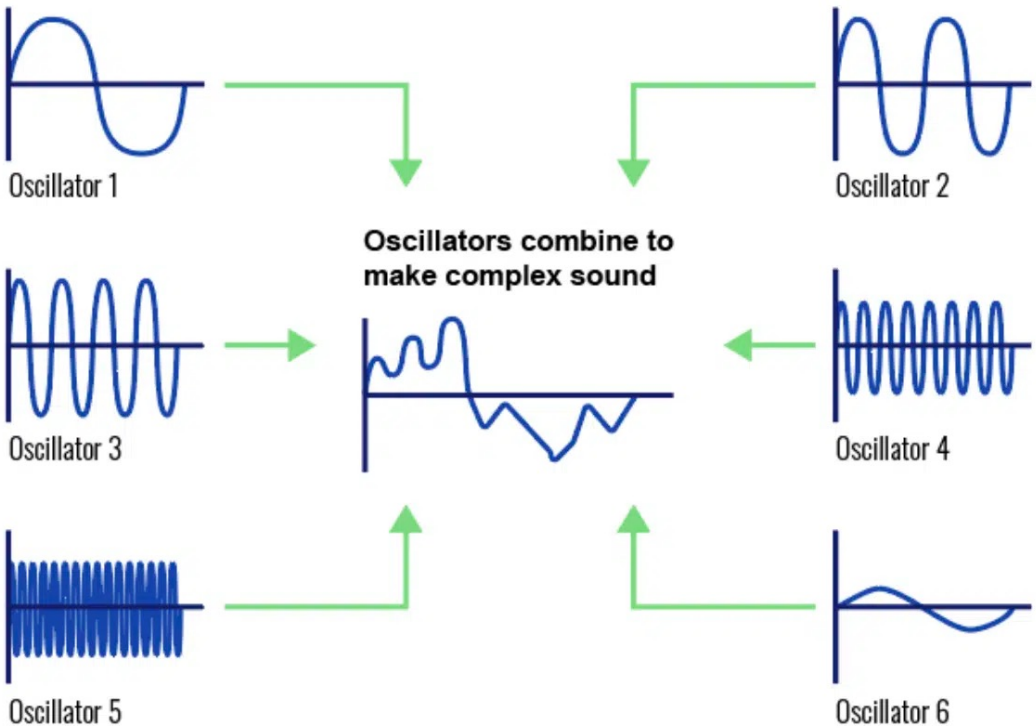
Spectrum


(a) Flute
0 50 100 150 200 250 300 350
Time (ms)
0 2000 4000 6000 8000 10000
Frequency (Hz)
(b) Trumpet
0 50 100 150 200 250 300 350
Time (ms)
0 2000 4000 6000 8000 10000
Frequency (Hz)
(c) Violin
0 50 100 150 200 250 300 350
Time (ms)
0 2000 4000 6000 8000 10000
Frequency (Hz)
(d) Flat-spectrum complex tone
0 50 100 150 200 250 300 350
Time (ms)
0 2000 4000 6000 8000 10000
Frequency (Hz)

Additive Synthesis

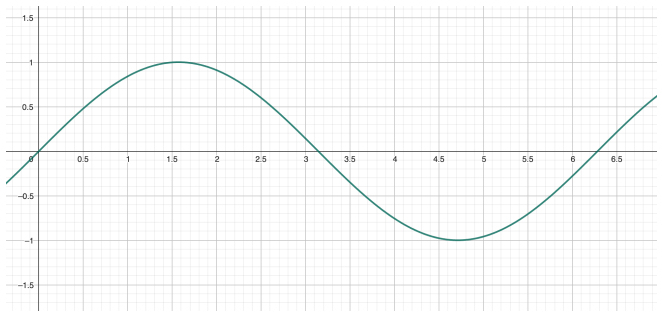
Additive Synthesis

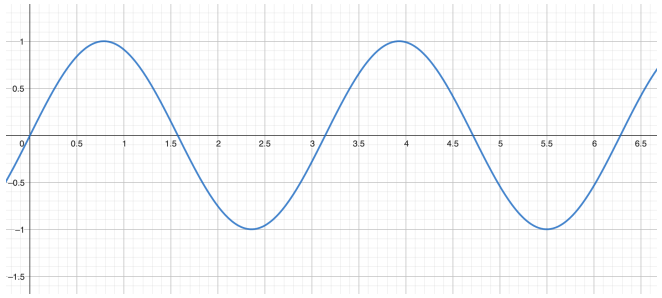
- Summation of sinusoids to create complex tones

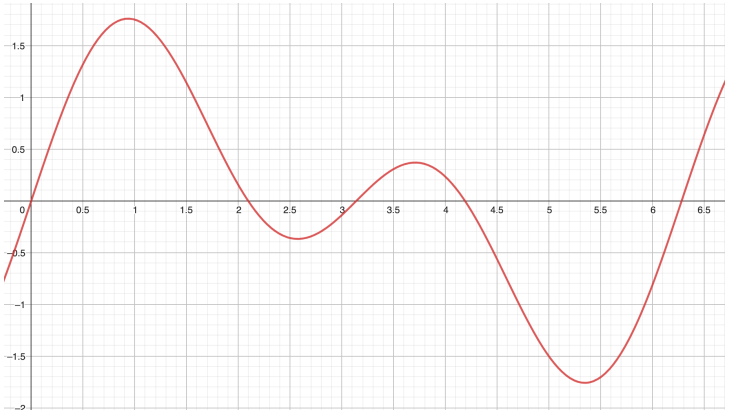


Additive Synthesis

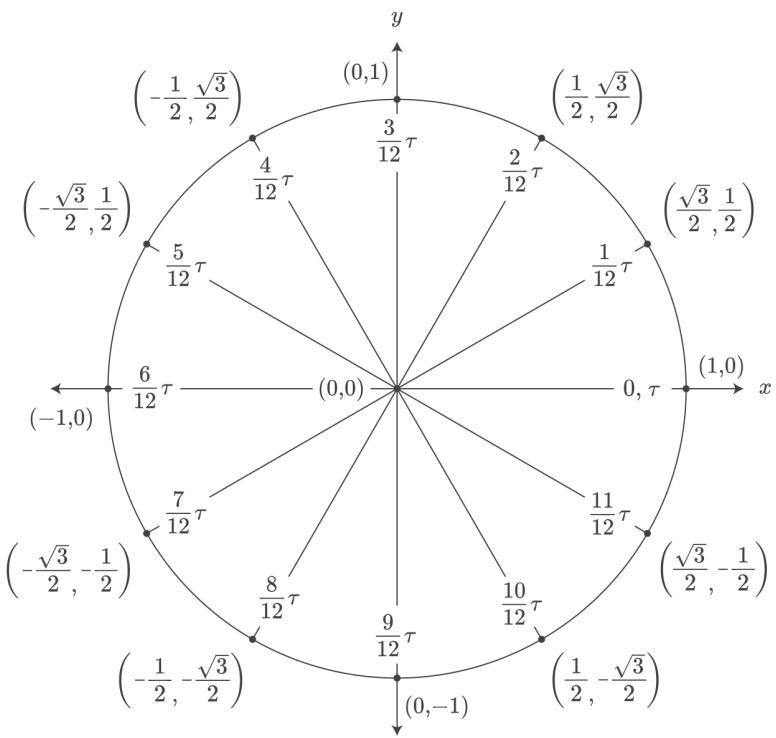
- Constructive and Destructive interference

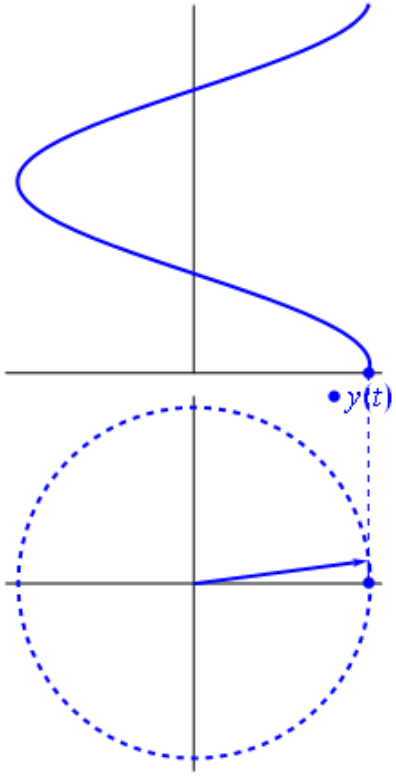






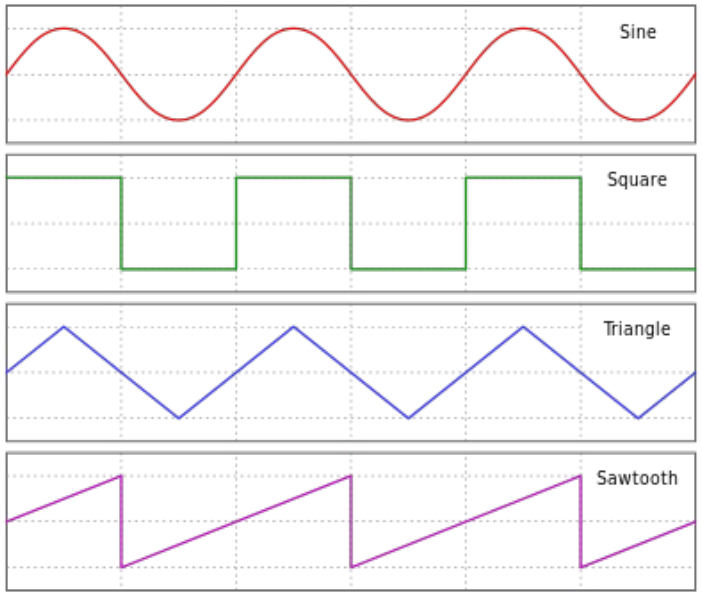
Aside: Picture phase on the unit circle





Additive Synthesis

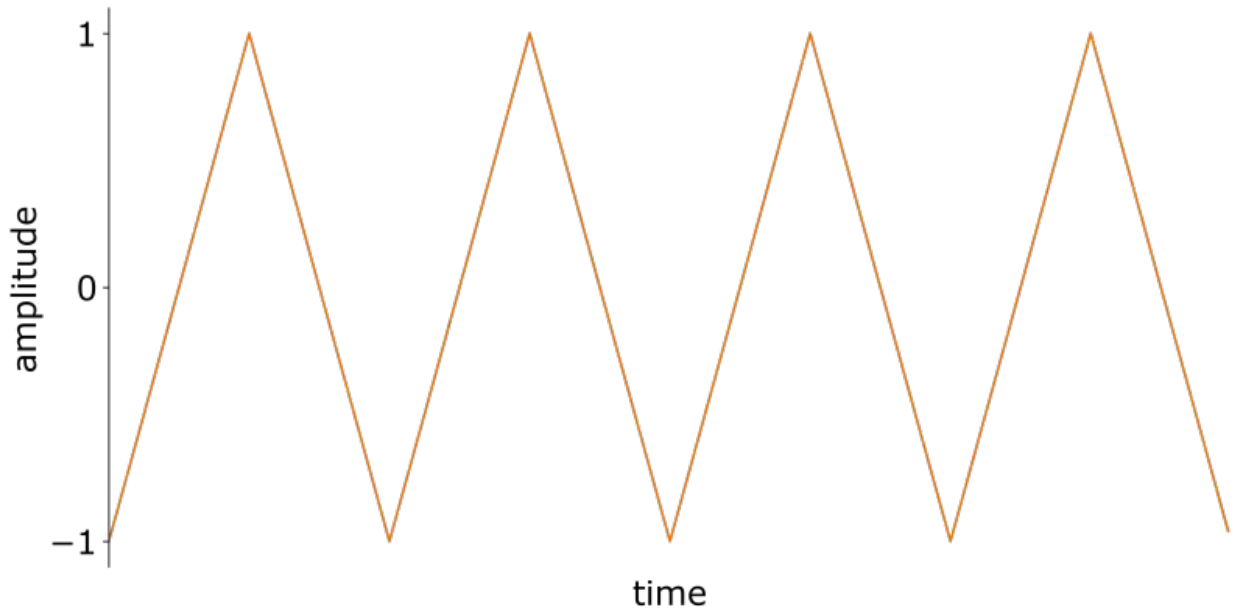
- We can construct harmonic and inharmonic sounds via additive synthesis
  - Harmonic – partials have integer ratio relationships
  - Inharmonic – no integer relationship between partials



Classical Waveforms

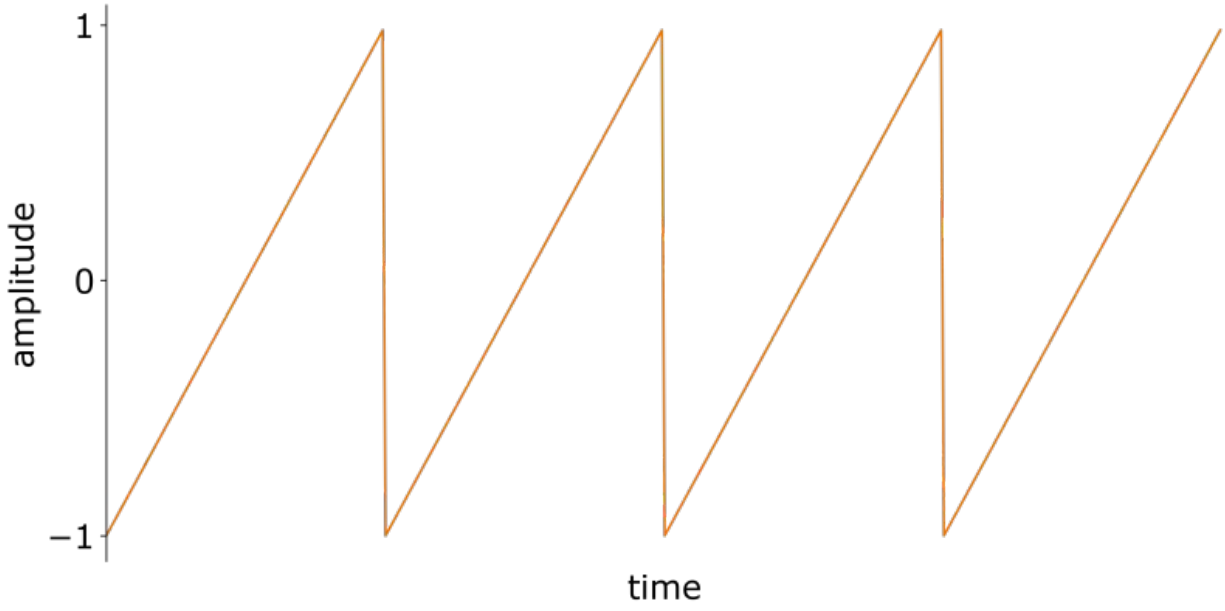
Triangle Wave

- Odd harmonics only at $\frac{1}{n^2}$ relative strength



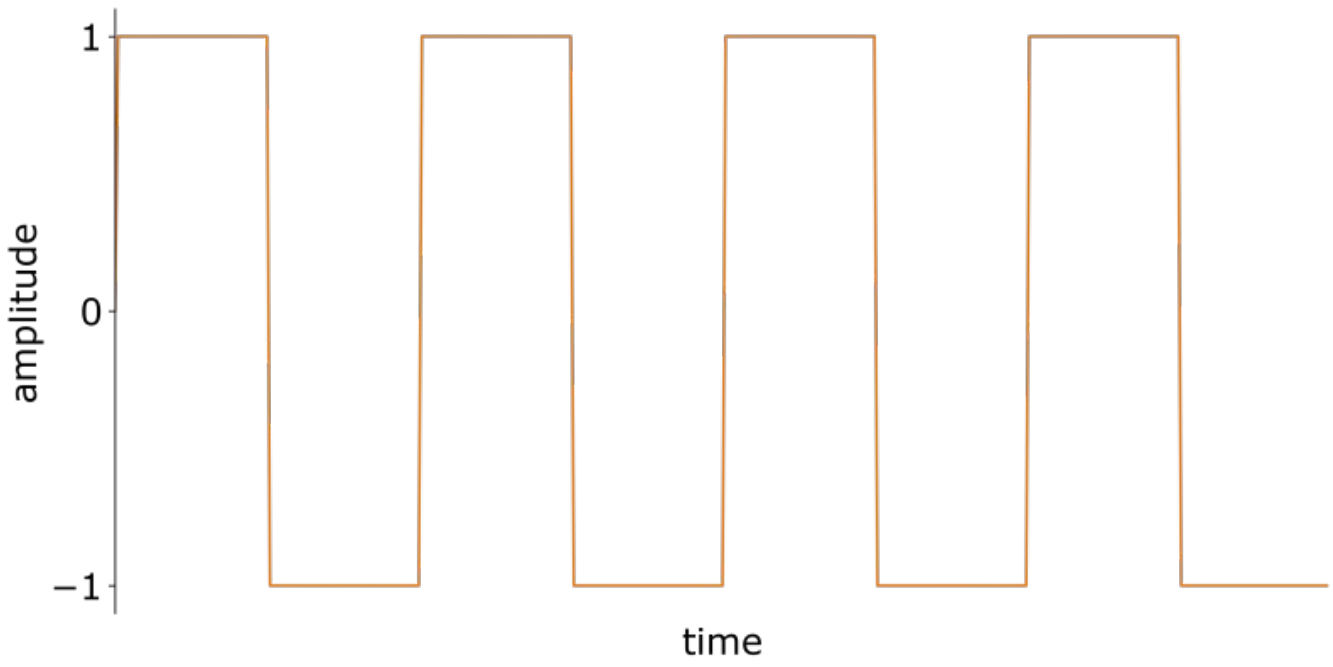
Sawtooth Wave

- All harmonics at $\frac{1}{n}$ relative strength



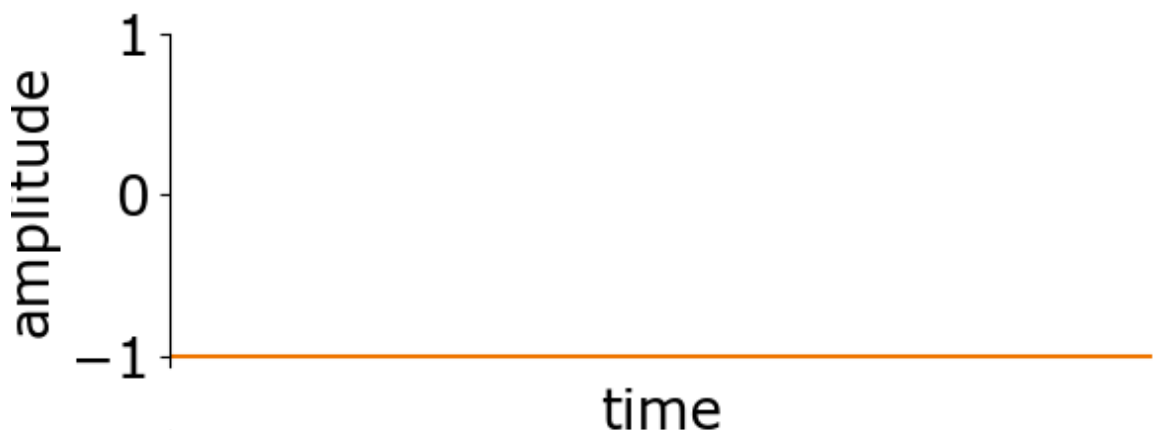
Square Wave

- Only odd harmonics at $\frac{1}{n}$ relative strength



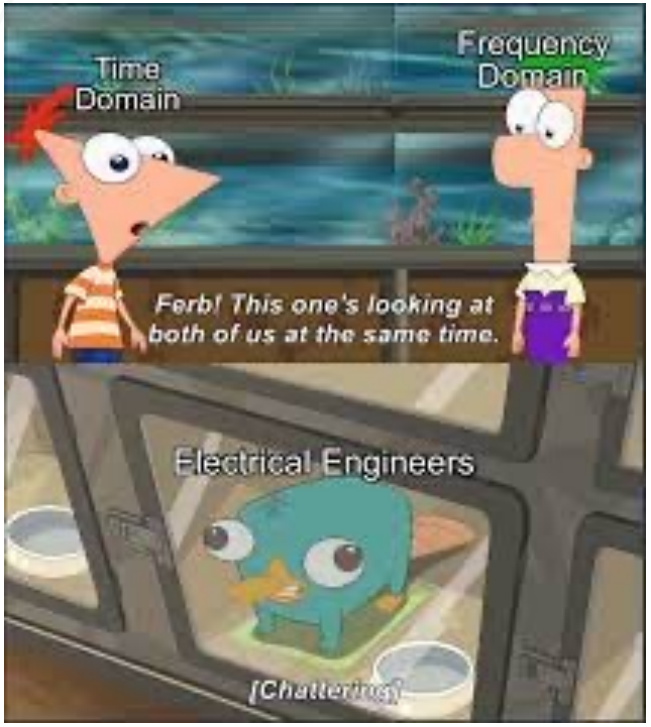
Pulse Train / Rectangle Wave

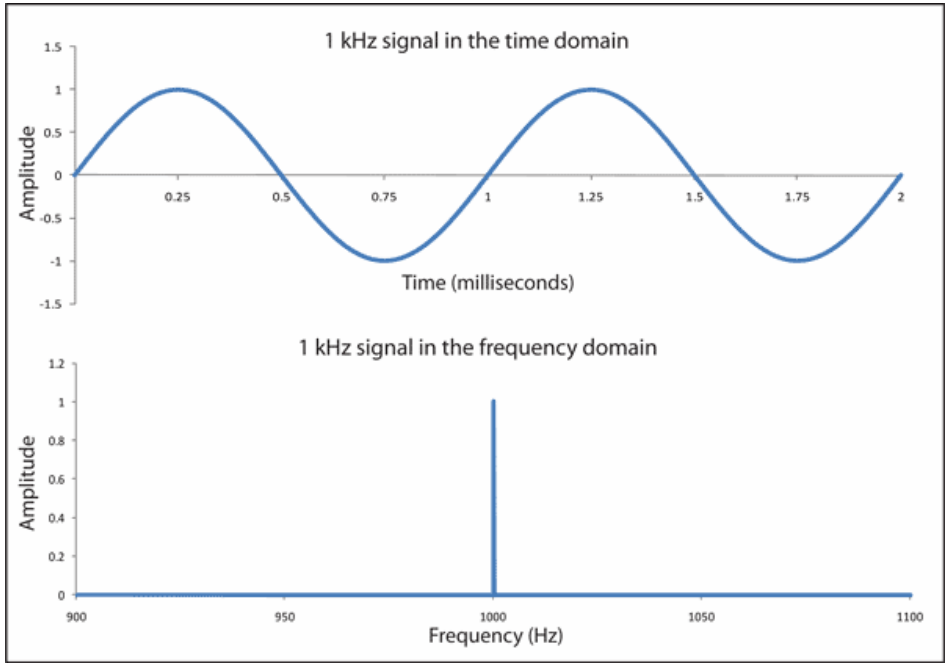
- General form of square wave
- Duty cycle
  - How much of the waveform's cycle is positive vs negative
  - Square wave duty cycle is 0.5 (half the time positive, half negative)
  - Change what partials are present based on ratio of length of shortest phase / length of cycle



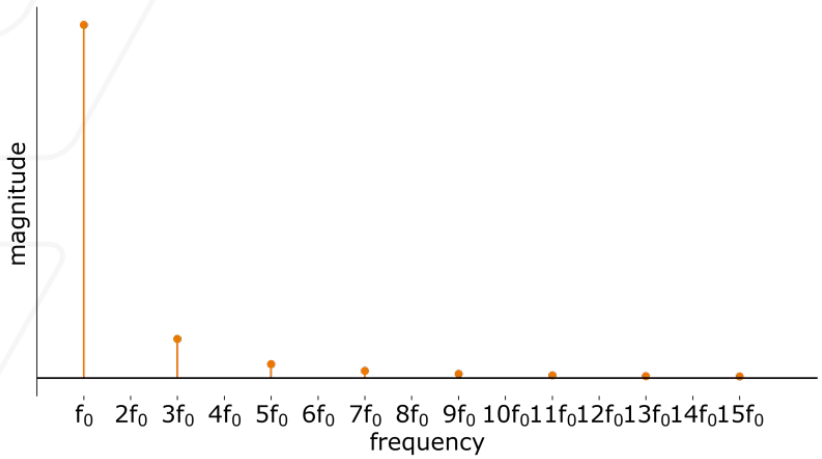
Remember: Magnitude Spectrum

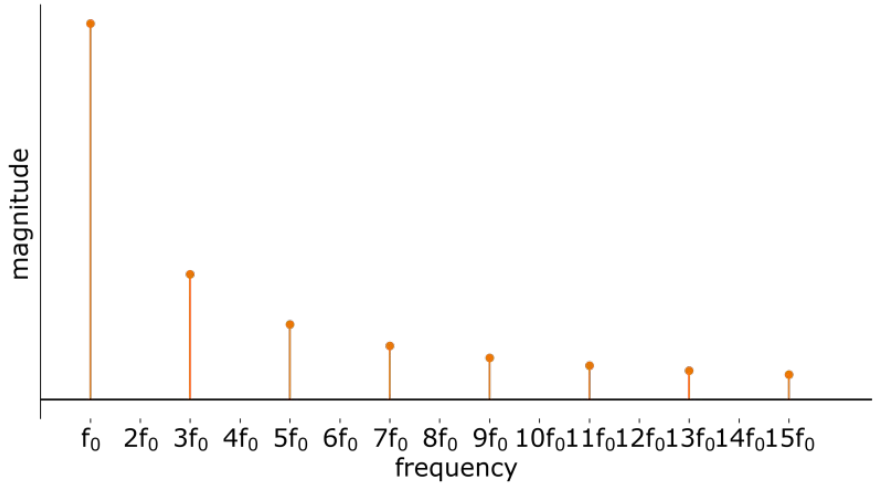
- How do we understand the frequency content of a signal?

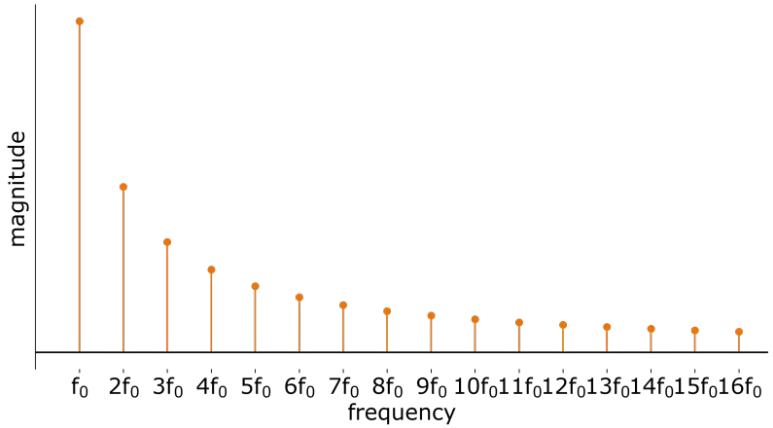




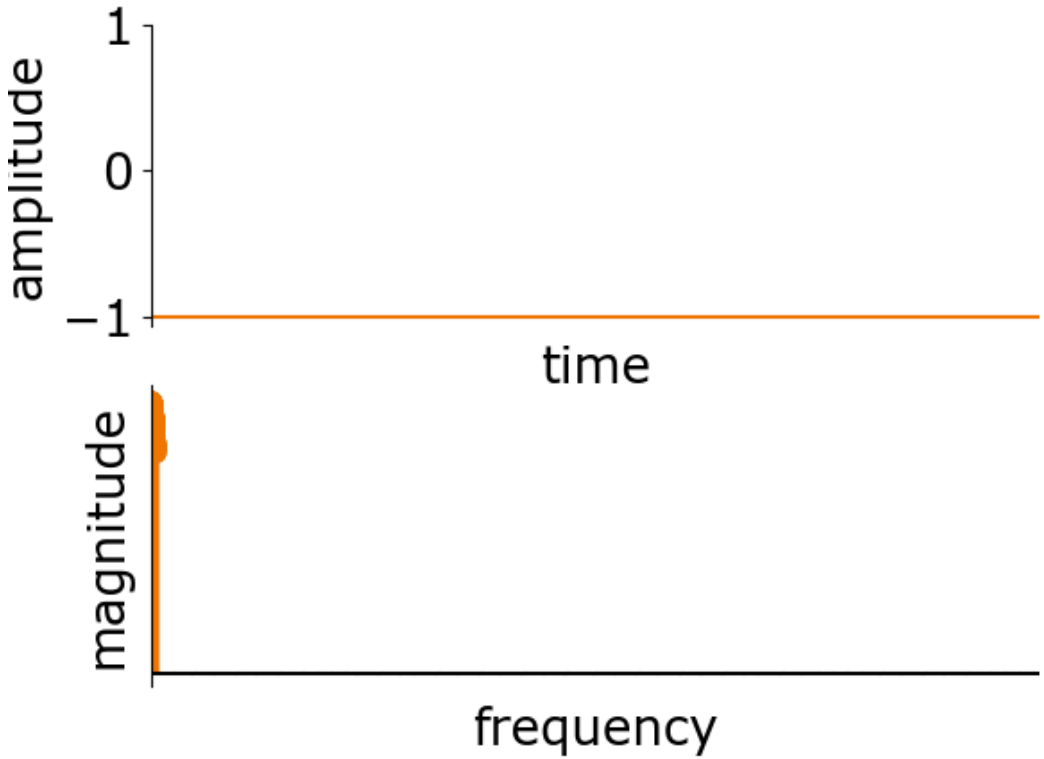
Classical Waveforms in the Frequency Domain





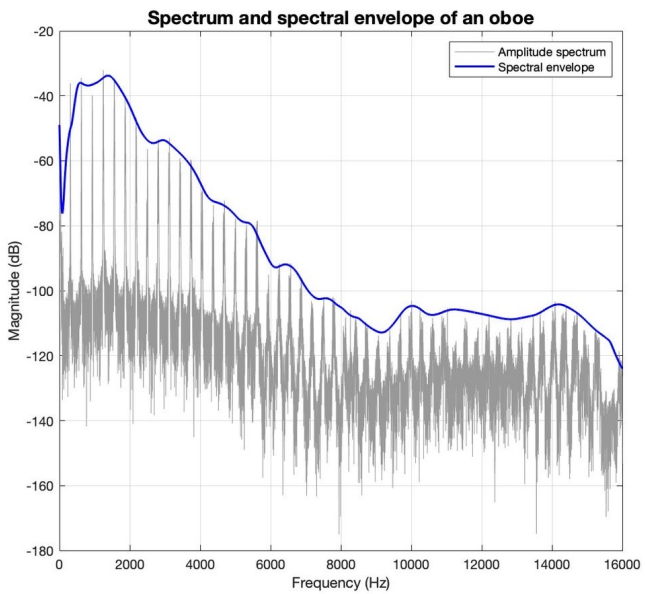


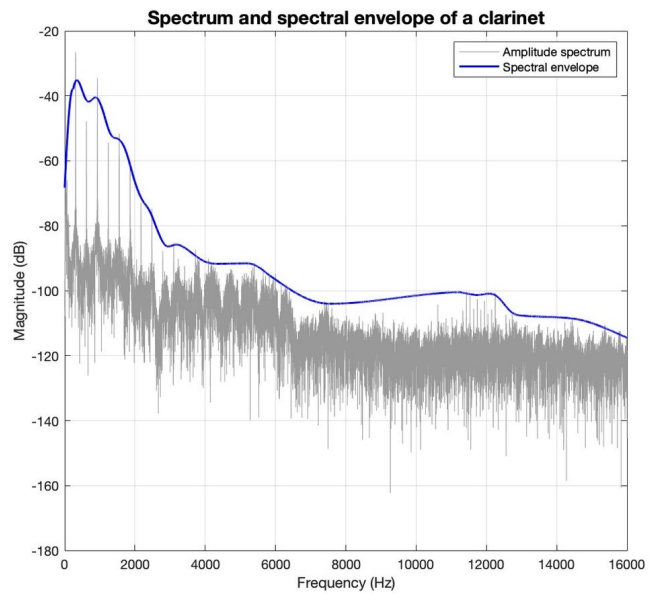
Classical Waveforms in the Frequency Domain



Spectral Envelope

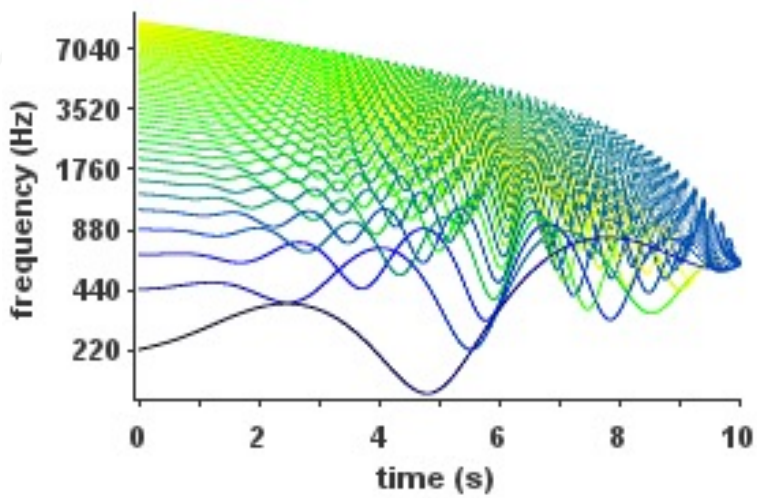
- A line over the peaks of the spectrum
- Represents the amplitude of partials in a signal

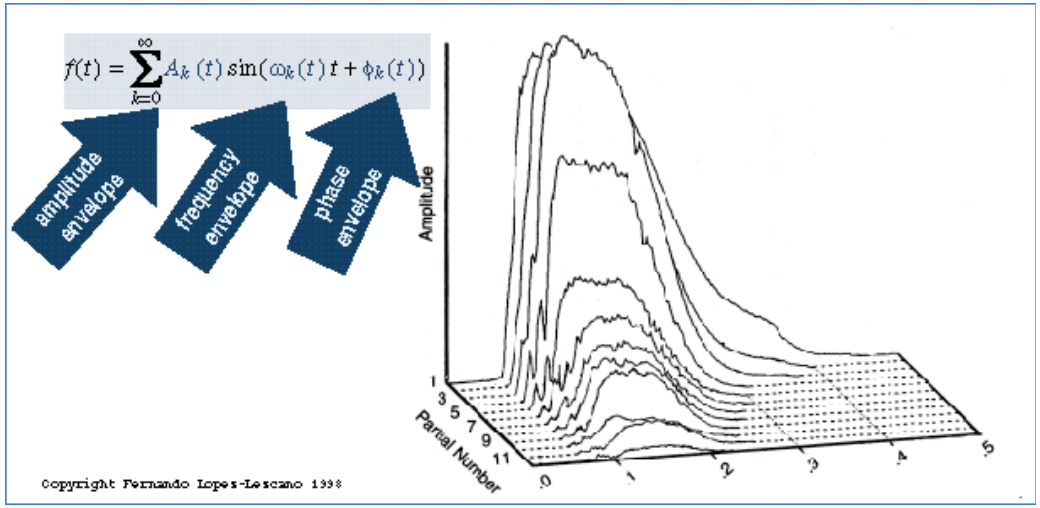




Time-varying additive synthesis

- Acoustic instruments have variations in harmonics over time
- We can vary the presence of harmonic partials over time to recreate organic sounds





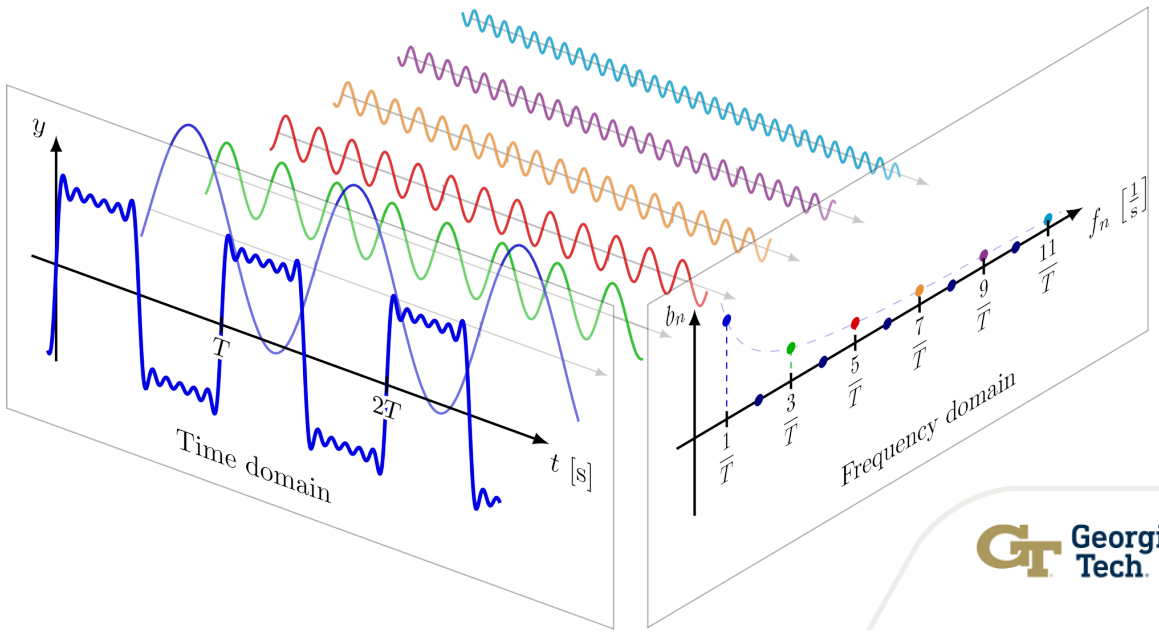
Time-varying additive synthesis

- In practice, this is extremely difficult
- Many frequencies present, all with unique amplitude values, changing at different time scales
- This is why timbre is so difficult to define (and manipulate)

- Strategies for Variable Additive Synthesis:
  - Frequency masking – trace the amplitude changes of the highest and lowest frequencies and adjust those in the middle accordingly
  - Interpolate between two or more spectral envelopes to create changes

Analysis Based Synthesis

- Fourier Synthesis
- Analyze the spectral content of an actual sound and use that as input to our additive synthesis
- More on this later…



Additive Synthesis in Max

Adding Signals

- +~ combines two signals
- If you put two signals into the same inlet, they will also be added
- Remember
  - When we add two signals, there amplitudes add (constructive interference)
  - If you do not scale the total amplitude, you will get clipping!

$$(a_1\cos(\omega t) + \emptyset) + (a_2\cos(\omega t) + \emptyset) = (a_1 + a_2)\cos(\omega t) + \emptyset$$

*given equal phase and frequency

Building Complex Waveforms in Max

- Work in groups to create:
  - Sawtooth
    - All harmonics at $\frac{1}{n}$ relative strength
  - Triangle
    - Odd harmonics only at $\frac{1}{n^2}$ relative strength
  - Square
    - Only odd harmonics at $\frac{1}{n}$ relative strength

- Make sure you can easily adjust the fundamental frequency
- Create up to 7 harmonics

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Up Next…

- Readings and Supplements
- Assignment 1 due on Friday at 11:59p